Part-time Penalties, Household Decisions, and the Gender Pay Gap

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Motivation

Female labor market outcomes are broadly shaped by:

- Unequal wages relative to male and occupational opportunities.

Neumark et al 1996, Goldin and Rouse 2000

- A disproportionate share of home production.

Cardia and Gomme 2018, Kleven et al 2019

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Within married households, labor supply is not an individual decision alone.

These forces interact with the **part-time penalty**, amplifying gender gaps in hours and wages.

French 2005, Goldin 2014, Bick et al. 2022, Erosa et al 2022

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Research question: How do parental-time allocation, female wage wedges, and nonlinear pay jointly shape gender gaps in hours, wages, and earnings?

What We Do

1. **Document empirical facts on time allocation and joint labor supply**
   - Child rearing drives the gender gap in home production.
   - Within couples, larger wage gaps map into larger hours gaps.

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- Within couples, larger wage gaps map into larger hours gaps.

2. Build a static household model

- Couples jointly choose labor supply, child rearing time/expenditure, and leisure.
- Females face lower labor-market returns and greater child-rearing weight.
- A part-time penalty links hours choices to wage outcomes.

3. Quantify mechanisms and policies

- Decompose the sources of gender gaps
- Evaluate policy implications of tax reforms

Preview of the results

1. Unequal weight in child rearing between males and females
   - accounts for most of the gender gap in hours
2. Labor market wedge accounts for most of the gender gap in wages
   - part-time penalty further amplifies this wage gap
3. Policy implications: switching from joint filing (e.g. the US) to individual filing (e.g. Sweden)
   - Increases the female employment rate by about 3 percentage points, slightly more for mothers
   - Women spend less time on child rearing and households substitute expenditure for child-rearing time

Empirical Facts

Data Sources

Sample: Married prime-age couples (25-54) with male employed

American Time Use Survey (ATUS)

- Detailed time allocation: market work, home production, child rearing, leisure, etc.
- ATUS is individual-level, so spouses are not observed jointly.

ACS Linked Married Couples 2015-2019

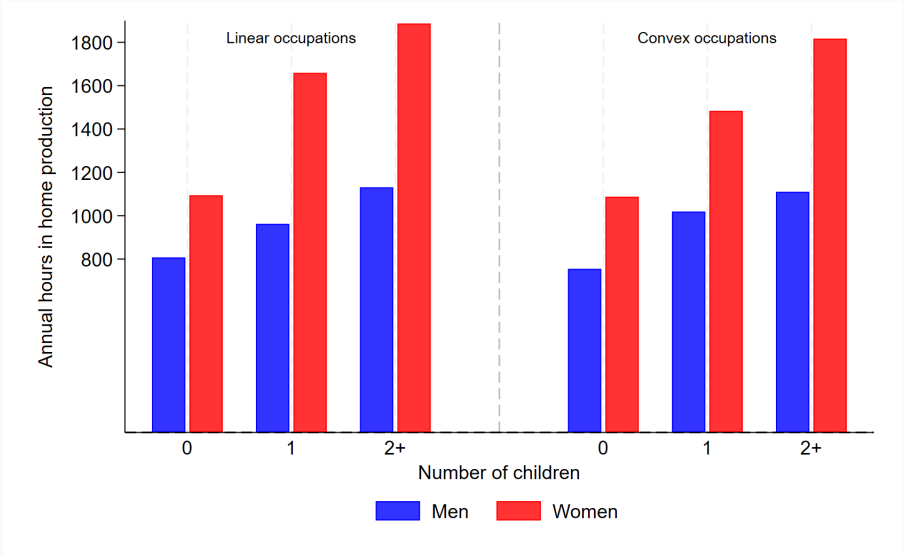
- Household-level data linking husbands and wives
- Joint labor market outcomes: hours, wages and participation.

Categorize occupations into two groups as linear and convex occupation.

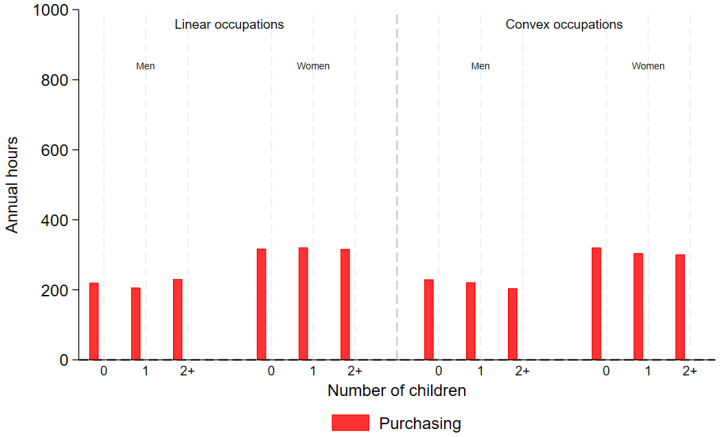
Classification of Occupations

- Divide occupations into 25 broad groups based on ACS classification.
- Estimate occupation-specific log hourly wages across 300-hour bins, controlling for worker characteristics.
- Calculate the employment-weighted slope of each occupation's wage-hour profile relative to 2,080–2,380 annual hours.
- Split occupations at the employment-weighted median slope (0.0605):
  - **Linear:** Below-median wage-hour slope (14 occupations).
  - **Convex:** Above-median wage-hour slope (11 occupations).
- Convex occupations account for 53.3% of employment.

Home Production Time by Gender and Kids

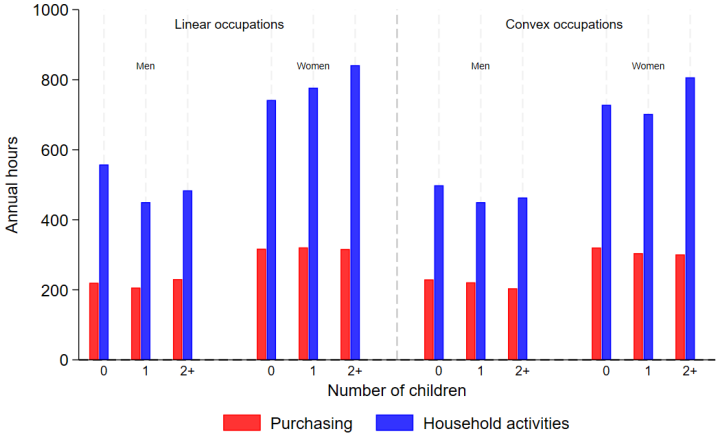


Anatomy of Home Production Time



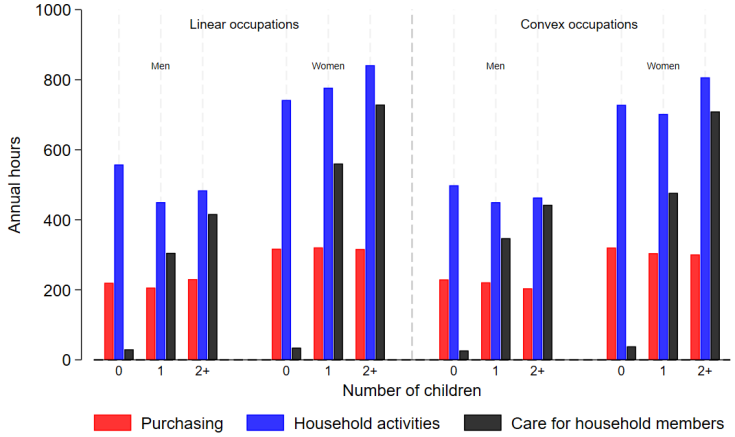
Household activities include housework, food preparation, cleaning, household management, garden care, animals and pets, vehicles, appliances, interior and exterior maintenance, decoration etc.

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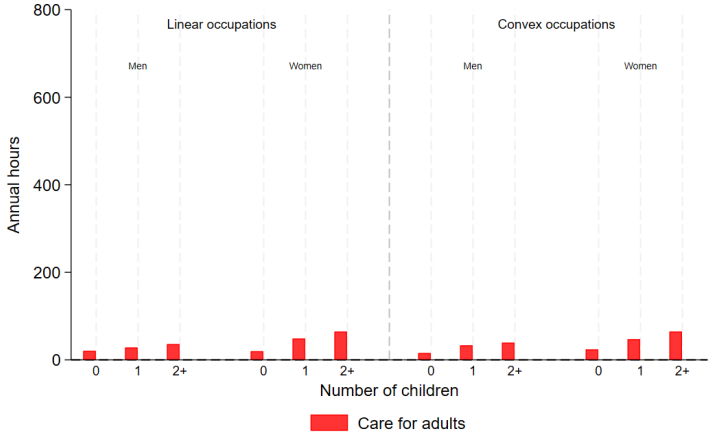
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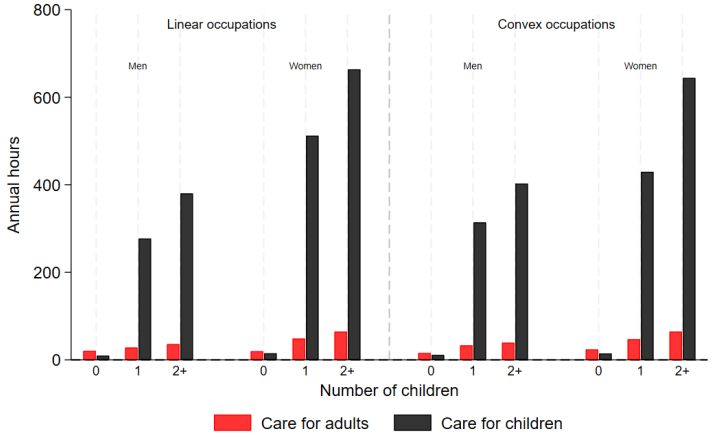


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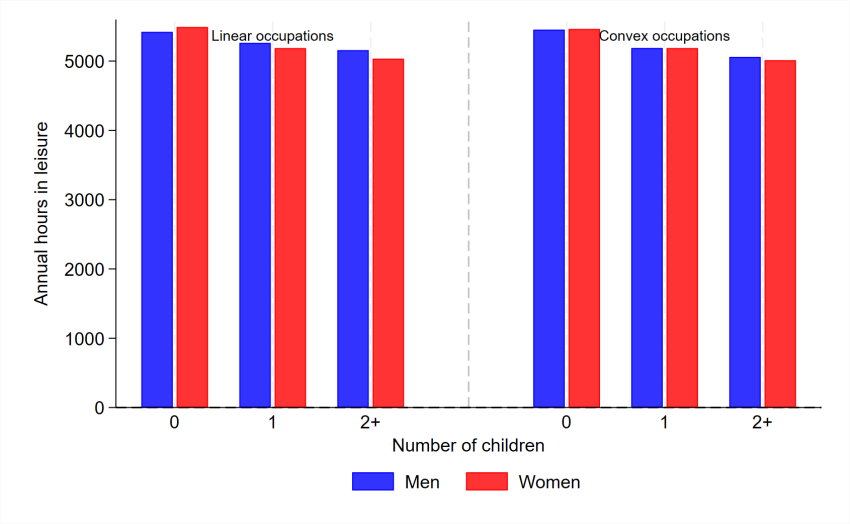
Care for Adults or Kids?



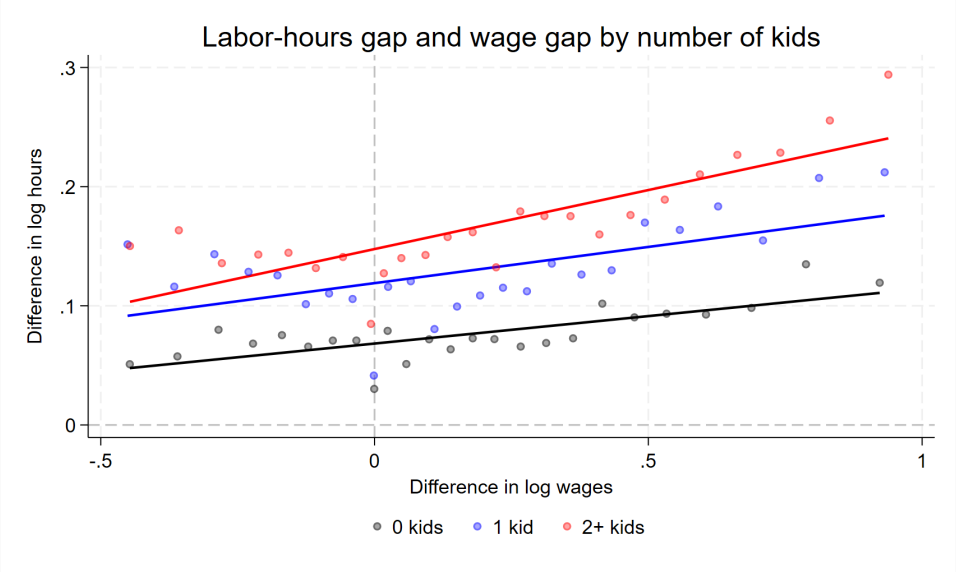
Care for Adults or Kids?



Similar Leisure Time between Male and Female



Joint labor supply: intensive margin



Joint labor supply: female employment rate

Husband occupation	0 kids	1 kid	2+ kids
Linear	0.82	0.72	0.59
Convex	0.81	0.70	0.59
Overall	0.81	0.71	0.59

Taking Stock

- Huge variation in home production time by gender and the number of kids
- Most of the variation comes from time spent on caring for children
- Leisure time is almost the same
- Children matter for both extensive and intensive margin of joint labor supply

Next step: build up a static intra-household model to quantify the role of child rearing

Model

Environment

A unit mass of married households in the economy. Each household consists of two members: male (m) and female (f). Spouses differ by

- their potential wages in the two occupations, linear (L) and nonlinear (NL):

$$\left(w_g^L, w_g^{NL}\right), \qquad g \in \{m, f\},$$

  jointly lognormal, with general ability correlated within the couple (ρ)
- their taste for leisure: $\Psi_{g,i}$
- the number of kids: n
- taste for child rearing: ξ

Fertility is exogenous while occupations are chosen jointly by the couple.

Household's problem

Given occupations (o_m, o_f), a household jointly chooses labor supply l_m, l_f, child rearing time q_m, q_f, child rearing expenditure e and leisure time h_m, h_f

$$
\begin{aligned}
&V\left(w_m^L, w_m^{NL}, w_f^L, w_f^{NL}, n, \xi, \Psi_{m,i}, \Psi_{f,i} \mid o_m, o_f\right) \\
&\quad = \max_{l_m, l_f, h_m, h_f, q_m, q_f, e} \alpha_c \log C(l_m, l_f, e) + (1-\alpha_c)\,\xi \lambda_n \log I(q_m, q_f, e) \\
&\qquad - \sum_{g \in \{m,f\}} \Psi_{g,i} \frac{(1-h_g)^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} - \frac{\chi_h}{2}(h_m - h_f)^2 \\
&\text{s.t.} \qquad l_m + q_m + h_m = 1, \qquad l_f + q_f + h_f = 1
\end{aligned}
$$

- $C(l_m, l_f, e)$: utility from joint consumption
- $I(q_m, q_f, e)$: utility from child rearing, adjusted by the number of kids λ_n
- χ_h: complementarity in spouses' leisure

Family income and consumption

Individuals face non-linear wage schedule

Bick, Blandin, Rogerson 2022

$$E(l \mid o) = \begin{cases} l^{1+\theta_o^B} & \text{if} \quad l < \bar{l} \\ B_o \cdot l^{1+\theta^A} & \text{if} \quad l \geq \bar{l} \end{cases}$$

- $\bar{l}$: full-time threshold
- θ_B: part-time penalty. θ_A: over-time penalty
- B_o adjust for discontinuity

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Family joint consumption

$$C(l_m, l_f, e) = w_m^{o_m} \cdot E(l_m \mid o_m) + w_f^{o_f}(1 - \delta_{o_f}) \cdot E(l_f \mid o_f) - e - T(y)$$

- females face direct labor market wedge $\delta_o > 0$ (by occupation).
- y: household labor income;
- $T(y)$: progressive tax function

Child rearing

The utility of child rearing is a nested CES of time q_f, q_m and expenditure e:

$$I(q_f, q_m, e) = \left[\alpha_I \left(Q(q_f, q_m)\right)^{\frac{\gamma_I - 1}{\gamma_I}} + (1 - \alpha_I)\, (e)^{\frac{\gamma_I - 1}{\gamma_I}}\right]^{\frac{\gamma_I}{\gamma_I - 1}}$$

- Q: effective time input
- γ_I: elasticity of substitution between time and expenditure

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- Q: effective time input
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The effective time input :

$$Q(q_f, q_m) = \left[\alpha_q q_m^{\frac{\gamma_q - 1}{\gamma_q}} + (1 - \alpha_q) q_f^{\frac{\gamma_q - 1}{\gamma_q}}\right]^{\frac{\gamma_q}{\gamma_q - 1}}.$$

- γ_q is the elasticity of substitution between male and female time
- α_q captures the weight.

Occupation and employment choice

Couples also choose occupations and employment status jointly:

$$d_m \in \{L, NL\} \quad \text{and} \quad d_f \in \{N, L, NL\}$$

- females can choose non-employment (N), the linear (L) or the nonlinear (NL) occupation

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- females can choose non-employment (N), the linear (L) or the nonlinear (NL) occupation

For each regime $r = (d_m, d_f) \in \mathcal{R}$, the household solves optimization:

$$V_r\left(w_m^L, w_m^{NL}, w_f^L, w_f^{NL}, n, \xi, \Psi_{m,i}, \Psi_{f,i}\right) = V\left(w_m^L, w_m^{NL}, w_f^L, w_f^{NL}, n, \xi, \Psi_{m,i}, \Psi_{f,i} \mid d_m, d_f\right)$$

Households draw idiosyncratic taste shocks ϵ_r from Gumbel distribution for each possible regime and choose

$$r^* = \arg\max_{r \in \mathcal{R}} \{V_r - q\,\mathbb{1}\{d_f \neq N\} + \epsilon_r\}$$

- q: utility cost when both spouses work

Parameterization

Calibration Strategy

- Exogenous parameters
  - Tax progressivity, part-time penalties, Frisch elasticity
  - Distribution of the households of number of kids

- Endogenous Parameters
- We simulate 100,000 households and choose the parameters Θ to minimize

$$\sum_k \left(m_k^{model}(\Theta) - m_k^{data}\right)^2$$

- Moments: hours, wages, child rearing time and expenditure, occupation and employment shares, and gender gaps; the hours distribution; within-couple hours and wage gaps

Parameters chosen internally

Parameter	Meaning	Targeted moment	Data	Model
Wages				
μ_0	Ability level	Mean log wage, men with no kids	3.32	3.31
a_{NL}	NL ability premium	Men's NL − L log wage premium	0.29	0.28
σ_A	Ability dispersion	SD of log wage, L	0.62	0.62
σ_B	Extra NL dispersion	SD of log wage, NL	0.65	0.64
ρ	Ability correlation in the couple	Correlation of spouses' log wages	0.46	0.47
δ_L	Female wage wedge, L	Gender gap in log wages, L	0.08	0.07
δ_{NL}	Female wage wedge, NL	Gender gap in log wages, NL	0.30	0.29
Taste for leisure				
Ψ_m	Men's leisure weight	Mean log hours, men	7.72	7.73
Ψ_f	Women's leisure weight	Mean log hours, employed women	7.56	7.55
$\sigma_{\psi m}$	Men's core dispersion	Men within 1,780–2,680 hours (%)	83.9	81.0
p_m	Men's tail share	Men below 1,780 hours (%)	4.5	5.4
κ_m	Men's tail width	Men at 2,380 hours or more (%)	28.0	24.9
$\sigma_{\psi f}$	Women's core dispersion	Women within 1,780–2,680 hours (%)	80.8	78.6
p_f	Women's tail share	Women below 1,780 hours (%)	15.1	18.5
κ_f	Women's tail width	Women at 2,380 hours or more (%)	12.7	12.4

Parameters chosen internally

Parameter	Meaning	Targeted moment	Data	Model
Household and regime choice				
χ_h	Spousal leisure complementarity	Gender gap in log leisure, employed (ATUS)	0.003	0.007
q	Dual-work cost	Female employment rate (%)	66.6	67.4
σ_ϵ	Regime taste shocks	Share of L–L couples (%)	15.5	15.2
Child rearing				
λ_1	Weight on child rearing, 1 kid	Child expenditure share, 1 kid (%)	17.1	17.1
λ_2	Weight on child rearing, 2+ kids	Child expenditure share, 2+ kids (%)	26.9	26.9
α_I	Weight on time vs. money	Mothers' log child time	6.37	6.38
γ_I	Time–money substitution	Mothers' log child time, non-employed − employed	0.50	0.52
α_q	Weight on father's time	Log child time gap, fathers – mothers	−0.47	−0.46
γ_q	Father–mother time substitution	Log child time gap, fathers – mothers, NL − L, 2+ kids	0.12	0.10
σ_ξ	Dispersion of taste for children	Mean log hours, employed mothers with 1 kid	7.56	7.55
Taxes				
λ_y	Level of the tax function	Average marginal tax rate	0.37	0.40

Model Fit: Occupation Shares

Target	Data	Model
A. Joint occupation and employment shares		
Husband *L*, wife *L*	15.5%	15.2%
Husband *L*, wife *NL*	10.8%	15.5%
Husband *L*, wife non-employed	13.4%	13.7%
Husband *NL*, wife *L*	18.6%	18.0%
Husband *NL*, wife *NL*	21.6%	18.7%
Husband *NL*, wife non-employed	20.0%	18.9%

Model Fit: Mean log working hours

Male			Female		
Moment	Data	Model	Moment	Data	Model
Linear occupation			Linear occupation		
0 child	7.68	7.73	0 child	7.59	7.64
1 child	7.70	7.71	1 child	7.52	7.53
2 children	7.71	7.71	2 children	7.48	7.48
Nonlinear occupation			Nonlinear occupation		
0 child	7.72	7.75	0 child	7.65	7.66
1 child	7.72	7.73	1 child	7.60	7.56
2 children	7.73	7.73	2 children	7.57	7.53

Model Fit: Gender Gaps

	Earnings		Hours		Hourly wage	
Gender gap (log)	Data	Model	Data	Model	Data	Model
By number of children						
0 child	0.24	0.27	0.08	0.09	0.16	0.17
1 child	0.34	0.36	0.15	0.17	0.19	0.19
2 children	0.46	0.43	0.20	0.22	0.26	0.21
By occupation						
Linear occupation	0.26	0.26	0.18	0.19	0.08	0.07
Nonlinear occupation	0.43	0.46	0.13	0.17	0.30	0.29
All couples	0.38	0.38	0.16	0.18	0.22	0.20

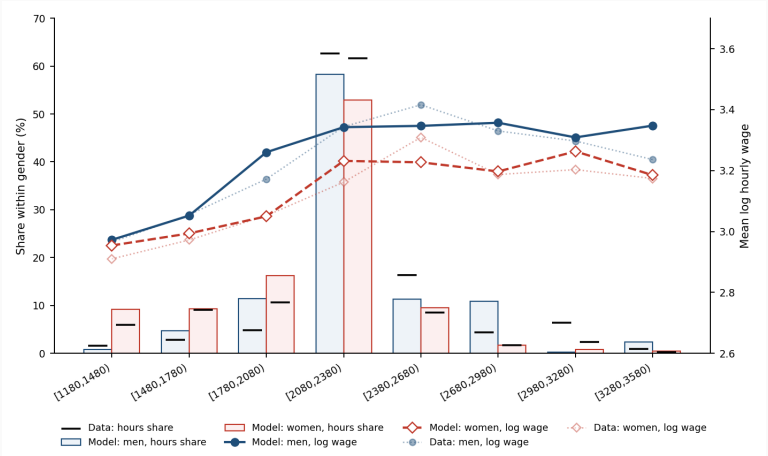
Gender gaps are male minus female mean logs.

Model Fit: Other

Log mean child rearing time					
Linear occupation	Data	Model	Nonlinear occupation	Data	Model
Male, 1 child	5.77	5.71	Male, 1 child	5.94	5.74
Male, 2 children	5.92	6.08	Male, 2 children	6.00	6.14
Female, 1 child	6.41	6.20	Female, 1 child	6.18	6.15
Female, 2 children	6.48	6.60	Female, 2 children	6.43	6.56

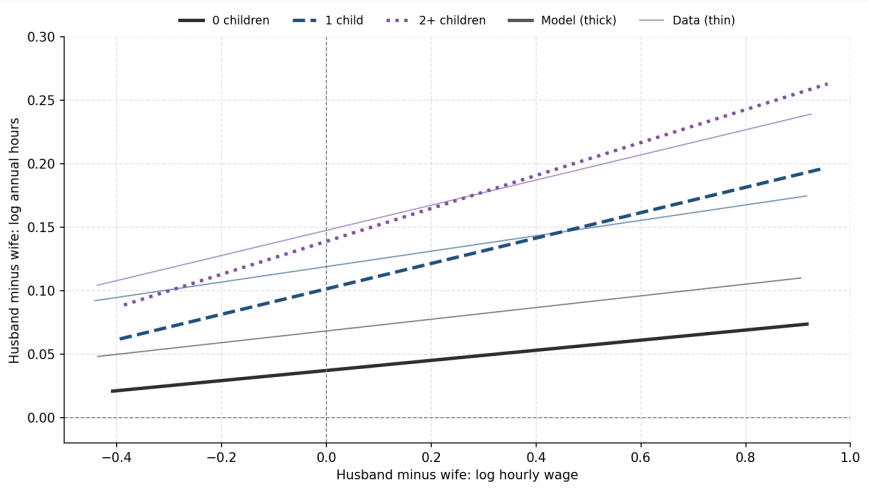
Moment	Data	Model
Avg. child expenditure/HH income, 1 child	17.1%	17.1%
Avg. child expenditure/HH income, 2 children	26.9%	26.9%
Female employment rate	66.6%	67.4%

Model Fit: Hours Distribution and Wages by Hours



Bars: share of workers by annual hours (model); black ticks: data. Lines: mean log hourly wage by hours bin, model and data (ACS).

Model Fit: Joint Labor Supply



Quantitative Analysis

Sources of Gender Gaps

Three sources of gender gaps in the model:

Gaps (in log)	Benchmark
Earnings	0.378
Wages	0.198
Hours	0.180

Sources of Gender Gaps

Three sources of gender gaps in the model:

- Unequal parental-time weights: $\alpha_q < 0.5$
  $\longrightarrow$ Women spend more time on child rearing.

Gaps (in log)	Benchmark	Equal weight
Earnings	0.378	0.255
Wages	0.198	0.173
Hours	0.180	0.083

Sources of Gender Gaps

Three sources of gender gaps in the model:

- Unequal parental-time weights: $\alpha_q < 0.5$
  $\longrightarrow$ Women spend more time on child rearing.
- Female wage wedge: $\delta_L, \delta_{NL} > 0$
  $\longrightarrow$ Women receive lower hourly wage rates.

Gaps (in log)	Benchmark	Equal weight	No wedge
Earnings	0.378	0.255	0.171
Wages	0.198	0.173	0.011
Hours	0.180	0.083	0.160

Sources of Gender Gaps

Three sources of gender gaps in the model:

- Unequal parental-time weights: $\alpha_q < 0.5$
  $\longrightarrow$ Women spend more time on child rearing.
- Female wage wedge: $\delta_L, \delta_{NL} > 0$
  $\longrightarrow$ Women receive lower hourly wage rates.
- Part-time penalty: $\theta_o^B > 0$
  $\longrightarrow$ Further amplifies the gaps

Gaps (in log)	Benchmark	Equal weight	No wedge	No part-time penalty
Earnings	0.378	0.255	0.171	0.366
Wages	0.198	0.173	0.011	0.146
Hours	0.180	0.083	0.160	0.220

Shapley Decomposition of Gender Gap

Shapley decomposition: attributes the total gap reduction to each channel in an additive way.

Gaps (log)	Equal weight	No wedge	No part-time penalty	Total reduction
Earnings	-0.123	**-0.204**	-0.013	-0.340
	(36.3%)	**(60.0%)**	(3.7%)	
Wages	-0.013	**-0.185**	**-0.037**	-0.236
	(5.7%)	**(78.4%)**	**(15.9%)**	
Hours	**-0.110**	-0.019	0.025	-0.105
	(105.1%)	(18.6%)	(-23.7%)	

- The wedge and the part-time penalty account for 94% of the wage gap
- Unequal parental-time weights account for essentially all of the gap in hours

Policy Implications

Joint filing and separate filing

What if married couples file taxes separately? $T(y) \to T(y_m) + T(y_f)$ (revenue neutral)

Joint filing and separate filing

What if married couples file taxes separately? $T(y) \to T(y_m) + T(y_f)$ (revenue neutral)

	Joint Filing	Separate Filing (revenue neutral)
	Gender gaps (in log)	
Earnings	0.378	0.363
Wages	0.198	0.195
Hours	0.180	0.168
	Female employment rate	
0 child	0.68	0.71
1 child	0.67	0.71
2 children	0.67	0.71

- Small reduction in gender hours/wages gap.
- Moderate increase in female employment rate (about 3 percentage points).

Joint filing and separate filing

	Joint Filing	Separate Filing
Avg child rearing time		
Male, 1 child	306.7	302.0
Male, 2 children	450.2	441.2
Female, 1 child	609.3	588.2
Female, 2 children	872.5	842.9
Avg child rearing expenditure (household)		
1 child	1.76	1.79
2 children	2.74	2.78

- Household increase labor supply $\rightarrow$ substitute child rearing time with expenditures

Conclusion

Conclusion

Gender gaps are shaped jointly by household decisions and the female wage wedge

- Child rearing is the main source of gender differences in hours.
- The wage wedge and the part-time penalty are the main sources of the wage gap.

Tax policy can increase female labor supply, but only partially closes gender gaps

- Separate filing raises female employment by about 3 percentage points and shifts households toward more market work and more expenditure-based child rearing.

Future work:

- More empirical analysis on child rearing time and expenditure
- Policy implications (e.g., childcare subsidies and part-time work allowances)

Appendix

Contribution to the Literature

1. **Children and joint labor supply** Albanesi and Olivetti 2009, Adda et al 2017, Cardia and Gomme 2018, Kleven et al 2019

   - Existing studies examine how children affect labor-market outcomes, household time allocation, and spousal labor supply.
   - Document how the relationship between within-couple wage and hours gaps becomes stronger with the number of children, and use these patterns to discipline our model.

2. **Nonlinear pay and joint labor supply** Goldin 2014, Cortés and Pan 2019, Bick et al 2022, Erosa et al 2022

   - Existing studies examine nonlinear returns to hours, household responsibilities, and their implications for gender wage gaps.
   - Quantify the interaction between unequal parental-time weights and part-time penalties in a model of joint labor supply.

3. **Taxation of married couples** Guner et al 2012, Bick and Fuchs-Schündeln 2018, Gayle and Shephard 2019, Borella et al 2023

   - How taxation of married couples affects secondary earners' labor supply and household decisions.
   - Analyze how separate filing affects female employment, gender gaps, and the allocation of childcare time and expenditures in a joint model.

Occupation Classification: Linear vs. Convex

1. **Occupation groups**
   - Map ACS occupations into 25 broad groups, using single men and women aged 25–45.
2. **Estimate wage-hour profiles**
   - Divide annual working hours into 300-hour bins. Use 2,080–2,380 as the reference.
   - Regress log hourly wages on hours-bin indicators. Control for sex, age, education, race, and public-sector employment.
3. **Calculate occupation-specific slopes**
   - Use adjusted wage differences over 880–2,380 annual hours.
   - Estimate employment-weighted slopes: $\widehat{s_j} = \frac{\sum_{k=-4}^{0} \omega_{jk} k \widehat{\beta}_{jk}}{\sum_{k=-4}^{0} \omega_{jk} k^2}$
4. **Classify occupations**
   - Calculate the employment-weighted median slope.
   - Median cutoff: 0.060535.
   - Linear: slope below cutoff (14 occupations).
   - Convex: slope at or above cutoff (11 occupations).

Heterogeneous taste for leisure: fat-tailed shocks

Each spouse draws a mean-one shifter to the leisure weight, $\Psi_{g,i} = \Psi_g \, \omega_{g,i}$, from a scale mixture of two lognormals:

$$\log \omega_{g,i} \sim (1 - p_g)\,\mathcal{N}\!\left(-\tfrac{\sigma^2_{\psi g}}{2}, \sigma^2_{\psi g}\right) + p_g\,\mathcal{N}\!\left(-\tfrac{\kappa_g^2 \sigma^2_{\psi g}}{2}, \kappa_g^2 \sigma^2_{\psi g}\right), \qquad g \in \{m, f\}$$

- A share $1 - p_g$ has similar leisure preferences (narrow core, log-SD $\sigma_{\psi g}$)
- A share p_g has unusual preferences (tail $\kappa_g > 1$ times wider)
- Both components have mean one, so Ψ_g is the average leisure weight
- $p_g = 0$ or $\kappa_g = 1$: the standard lognormal

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Why: a narrow core lets many workers bunch at the full-time threshold, while the tail generates part-time work and long hours.

Key parameters: taste for leisure

	Men	Women
Average leisure weight Ψ_g	45.5	134.9
Core dispersion $\sigma_{\psi g}$	0.34	1.73
Share in the tail p_g	0.34	0.03
Tail width κ_g	3.10	1.98

- Men: two-thirds have similar leisure preferences $\longrightarrow$ 58% work full time (data 63%)
- Women: preferences are widely dispersed but close to lognormal; hours vary mostly with children
- Spousal complementarity $\chi_h = 22.8$; dual-work cost $q = 0.13$

Parameters chosen externally

Parameters	Meaning	Value
τ_y	Tax progressivity	0.18
$\theta_L^B,\ \theta_{NL}^B$	Part-time penalty (L, NL)	0.33, 0.652
θ^A	Over-time penalty	0
$\bar{l}$	Full-time threshold	2,080 hours
α_c	Weight on (log) consumption	0.5
η	Frisch elasticity of labor supply	$1/3$

Calibration Parameters I

Block	Parameter	Meaning	Value
A. Potential wage distribution			
Wage level	μ_0	Common level of potential log wages	3.6575
Nonlinear wage premium	a_{NL}	Mean potential-wage premium in the nonlinear occupation	0.6880
General ability	σ_A	Dispersion of ability common to both occupations	0.5880
Comparative advantage	σ_B	Dispersion of comparative advantage in the nonlinear occupation	0.0076
Assortative matching	ρ_A	Cross-spouse correlation in general ability	0.5786
Female wage wedge	δ_1	Female wage wedge in the linear occupation	0.0676
Female wage wedge	δ_2	Female wage wedge in the nonlinear occupation	0.2365
B. Labor-supply and employment preferences			
Male work disutility	Ψ_m	Male weight on disutility from non-leisure time	45.4825
Female work disutility	Ψ_f	Female weight on disutility from non-leisure time	134.8512
Male disutility shocks	$\sigma_{\psi m}$	Core log-SD of the mean-one shifter to Ψ_m	0.3350
Male shock tail share	p_m	Share of men drawing from the wide tail	0.3362
Male shock tail width	κ_m	Tail log-SD relative to $\sigma_{\psi m}$	3.0982
Female disutility shocks	$\sigma_{\psi f}$	Core log-SD of the mean-one shifter to Ψ_f	1.7264
Female shock tail share	p_f	Share of women drawing from the wide tail	0.0260
Female shock tail width	κ_f	Tail log-SD relative to $\sigma_{\psi f}$	1.9801
Dual-work cost	q	Utility cost in regimes where both spouses work	0.1266
Regime taste shock	σ_ϵ	Scale of Type-I extreme-value shocks over the six household regimes	0.4471
Spousal complementarity	χ_h	Complementarity in spouses' leisure	22.8267

Calibration Parameters II

Block	Parameter	Meaning	Value
C. Consumption and household resources			
Consumption weight	α_c	Weight on household consumption (fixed)	0.5000
Resource level	λ_y	Tax-schedule level; $1 - \lambda_y$ is the average tax rate at $\bar{Y}$	0.7039
D. Child-rearing technology and preferences			
Child-time substitution	γ_q	Elasticity of substitution between male and female child time	0.0858
Time–money substitution	γ_I	Elasticity of substitution between parental time and expenditure	0.7967
Time–money allocation	α_I	Weight on parental time relative to monetary child investment	0.2887
Parental-time allocation	α_q	Weight on father's contribution to parental childcare time	0.0010
Preference heterogeneity	σ_ξ	Dispersion in household tastes for child rearing	0.7135
Child-rearing multiplier	λ_1	Child-rearing multiplier for one-child households	0.4582
Child-rearing multiplier	λ_2	Child-rearing multiplier for two-child households	0.8657

Model Fit: Mean Log Annual Hours

Target	Data	Model
A. Linear occupation		
Mean log annual hours, male, *L*, 0 children	7.68	7.73
Mean log annual hours, female, *L*, 0 children	7.59	7.64
Mean log annual hours, male, *L*, 1 child	7.70	7.71
Mean log annual hours, female, *L*, 1 child	7.52	7.53
Mean log annual hours, male, *L*, 2 children	7.71	7.71
Mean log annual hours, female, *L*, 2 children	7.48	7.48
B. Nonlinear occupation		
Mean log annual hours, male, *NL*, 0 children	7.72	7.75
Mean log annual hours, female, *NL*, 0 children	7.65	7.66
Mean log annual hours, male, *NL*, 1 child	7.72	7.73
Mean log annual hours, female, *NL*, 1 child	7.60	7.56
Mean log annual hours, male, *NL*, 2 children	7.73	7.73
Mean log annual hours, female, *NL*, 2 children	7.57	7.53

Model Fit: Mean log hourly wages

Male			Female		
Moment	Data	Model	Moment	Data	Model
Linear occupation			Linear occupation		
0 child	3.15	3.17	0 child	3.12	3.10
1 child	3.20	3.16	1 child	3.14	3.10
2 children	3.21	3.16	2 children	3.10	3.08
Nonlinear occupation			Nonlinear occupation		
0 child	3.42	3.44	0 child	3.20	3.18
1 child	3.47	3.44	1 child	3.22	3.15
2 children	3.52	3.45	2 children	3.16	3.14

Model Fit: Overall Standard Deviations

Target	Data	Model
A. By occupation		
Standard deviation of log annual hours, L	0.31	0.25
Standard deviation of log hourly wages, L	0.62	0.62
Standard deviation of log annual hours, NL	0.24	0.22
Standard deviation of log hourly wages, NL	0.65	0.64
B. By gender		
Standard deviation of log annual hours, male	0.22	0.17
Standard deviation of log hourly wages, male	0.65	0.63
Standard deviation of log annual hours, female	0.31	0.27
Standard deviation of log hourly wages, female	0.62	0.63

Model Fit: Gender–Occupation Standard Deviations

Target	Data	Model
A. Male		
Standard deviation of log annual hours, male, *L*	0.26	0.18
Standard deviation of log hourly wages, male, *L*	0.63	0.61
Standard deviation of log annual hours, male, *NL*	0.20	0.16
Standard deviation of log hourly wages, male, *NL*	0.63	0.61
B. Female		
Standard deviation of log annual hours, female, *L*	0.33	0.28
Standard deviation of log hourly wages, female, *L*	0.61	0.62
Standard deviation of log annual hours, female, *NL*	0.27	0.26
Standard deviation of log hourly wages, female, *NL*	0.63	0.63

Model Fit: Childcare Time and Expenditure

Target	Data	Model
A. Log mean annual childcare hours: linear occupation		
Log mean annual childcare hours, male, *L*, 1 child	5.77	5.71
Log mean annual childcare hours, female, *L*, 1 child	6.41	6.20
Log mean annual childcare hours, male, *L*, 2 children	5.92	6.08
Log mean annual childcare hours, female, *L*, 2 children	6.48	6.60
B. Log mean annual childcare hours: nonlinear occupation		
Log mean annual childcare hours, male, *NL*, 1 child	5.94	5.74
Log mean annual childcare hours, female, *NL*, 1 child	6.18	6.15
Log mean annual childcare hours, male, *NL*, 2 children	6.00	6.14
Log mean annual childcare hours, female, *NL*, 2 children	6.43	6.56
C. Child expenditure		
Child expenditure relative to household net resources, 1 child	17.10%	17.13%
Child expenditure relative to household net resources, 2 children	26.90%	26.89%

Model Fit: Gender Gaps by Number of Children

Target	Data (Target)	Model
A. No children		
Gender gap in log annual earnings, 0 children	0.24	0.27
Gender gap in log annual hours, 0 children	0.08	0.09
Gender gap in log hourly wage, 0 children	0.16	0.17
B. One child		
Gender gap in log annual earnings, 1 child	0.34	0.36
Gender gap in log annual hours, 1 child	0.15	0.17
Gender gap in log hourly wage, 1 child	0.19	0.19
C. Two children		
Gender gap in log annual earnings, 2 children	0.46	0.43
Gender gap in log annual hours, 2 children	0.20	0.22
Gender gap in log hourly wage, 2 children	0.26	0.21

All gender gaps are male minus female, measured in log points.

Model Fit: Gender Gaps by Occupation and Overall

Target	Data (Target)	Model
A. Convex occupation		
Gender gap in log annual earnings, convex occupation	0.43	0.46
Gender gap in log annual hours, convex occupation	0.13	0.17
Gender gap in log hourly wage, convex occupation	0.30	0.29
B. Linear occupation		
Gender gap in log annual earnings, linear occupation	0.26	0.26
Gender gap in log annual hours, linear occupation	0.18	0.19
Gender gap in log hourly wage, linear occupation	0.08	0.07
C. Overall		
Gender gap in log annual earnings, all couples	0.38	0.38
Gender gap in log annual hours, all couples	0.16	0.18
Gender gap in log hourly wage, all couples	0.22	0.20

All gender gaps are male minus female, measured in log points.

Decomposing the No Part-time Penalty Channel

Effect on gaps	Wage gap	Hours gap	Earnings gap
Benchmark	0.198	0.180	0.378
Mechanical: new pay, benchmark hours and jobs	−0.052	0	−0.052
Responses: hours, occupation and participation	0	+0.040	+0.040
men's hours (same job)	0	−0.042	−0.042
women's hours (same job)	0	+0.075	+0.075
occupation switching	0	−0.002	−0.002
women's entry and exit	0	+0.008	+0.008
No part-time penalty, total	0.146	0.220	0.366

$\theta_o^B = 0$ with pay at 2,080 hours unchanged; mechanical = new pay schedule at benchmark hours and occupations. Gaps are male minus female, in log points. Households share the same random draws in every scenario, so each household is followed across runs. "Same job": employed in both scenarios in the same occupation. Response rows add up to the total change. Tax function held at its benchmark.

Equal Weight and the Part-time Penalty

Equalizing parental-time weights: $\alpha_q = 0.001$ to $\alpha_q = 0.5$.

Hourly wage gap	$\alpha_q = 0.001$	$\alpha_q = 0.5$	Reduction
With part-time penalty	0.198 (B)	0.173 (R)	0.025
No part-time penalty	0.146 (P)	0.145 (PR)	0.001

The additional effect of equalizing weights when the penalty is present:

$$A_R^W = (G_B^W - G_R^W) - (G_P^W - G_{PR}^W) = 0.025 - 0.001 = \mathbf{0.024}.$$

- The penalty amplifies the wage effect of unequal parental-time weights.
- Without the penalty, equalizing these weights has little effect on the wage gap.

G^W: male minus female mean log hourly wages. No part-time penalty: $\theta_o^B = 0$ below 2,080 hours, with pay at and above that threshold unchanged.

Equal Weight and the Part-time Penalty: Hours and Earnings

Reduction in gender gaps from equalizing parental-time weights:

	Wages	Hours	Earnings
With penalty: $G_B - G_R$	0.025	0.097	0.123
Without penalty: $G_P - G_{PR}$	0.001	0.126	0.126
Interaction: A_R	$+\mathbf{0.024}$	$-\mathbf{0.029}$	$-\mathbf{0.003}$

- Equalizing parental-time weights narrows the hours gap under both schedules, with a larger reduction without the penalty.
- The positive wage interaction and negative hours interaction largely offset.
- The earnings-gap reduction is almost unchanged across the two schedules.

Gaps are male minus female mean logs, in log units. Calculations use rounded reported gaps. The interactions include endogenous household responses and selection.

No Wedge and the Part-time Penalty

Removing the female wage wedge: δ_L, δ_{NL} to $\delta_L = \delta_{NL} = 0$.

Hourly wage gap	$\delta_o > 0$	$\delta_o = 0$	Reduction
With part-time penalty	0.198 (B)	0.011 (D)	0.187
No part-time penalty	0.146 (P)	-0.036 (PD)	0.182

The additional effect of removing the wedge when the penalty is present:

$$A_D^W = (G_B^W - G_D^W) - (G_P^W - G_{PD}^W) = 0.187 - 0.182 = \mathbf{0.005}.$$

- Removing the wedge closes almost all of the wage gap, with or without the penalty.
- The penalty barely changes this effect: the wedge lowers women's hourly rate directly, at every hours level.

G^W: male minus female mean log hourly wages. No part-time penalty: $\theta_o^B = 0$ below 2,080 hours, with pay at and above that threshold unchanged.

No Wedge and the Part-time Penalty: Hours and Earnings

Reduction in gender gaps from removing the wedge:

	Wages	Hours	Earnings
With penalty: $G_B - G_D$	0.187	0.020	0.207
Without penalty: $G_P - G_{PD}$	0.182	0.022	0.203
Interaction: A_D	$+\mathbf{0.005}$	$-\mathbf{0.002}$	$+\mathbf{0.004}$

- Removing the wedge narrows the hours gap only slightly (0.020 and 0.022).
- All three interactions are close to zero: the wedge and the part-time penalty act almost additively, unlike unequal parental-time weights, which work through hours.

Gaps are male minus female mean logs, in log units. Calculations use rounded reported gaps. The interactions include endogenous household responses and selection. With neither channel the wage gap turns negative, likely through selection into employment.

Joint filing and separate filing

	Joint Filing	Separate Filing
Avg child rearing time		
Male, 1 child	306.7	302.0
Male, 2 children	450.2	441.2
Female, 1 child	609.3	588.2
Female, 2 children	872.5	842.9
Avg log labor supply		
Male, 1 child	7.72	7.71
Male, 2 children	7.72	7.72
Female, 1 child	7.55	7.55
Female, 2 children	7.50	7.52
Average marginal tax rate (%)		
Aggregate (HH)	39.9	40.0
Aggregate (individual)	40.4	38.5
Male, 0 child	40.3	40.2
Male, 1 child	39.8	39.9
Male, 2 children	39.7	40.1
Female, 0 child	41.8	37.6
Female, 1 child	41.3	36.2
Female, 2 children	41.1	35.6
Avg child rearing expenditure (household)		
1 child	1.76	1.79
2 children	2.74	2.78